ECONOMETRICS I

Lecture 6

What variables to include in a regression?

Bias-efficiency tradeoffs

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Introduction

We want to answer:

- What variables should I include in a regression?
- All possible ones?
- What happens if an important variable is missing?
- What happens if an unimportant one is included?

Thus: perhaps the most important lecture!

Key insights:

- Tradeoff: bias vs. variance.
- Some tools to discriminate.

Remember our OLS-estimator formulas

In the **univariate** model ($\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x$), the **slope** estimate is

$$\boxed{\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)}}$$

and its **variance** in repeated sampling is

$$\boxed{\text{var}(\hat{\beta}_2 | x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}}$$

Remember our OLS-estimator formulas

In the **multivariate** case ($\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k$), the slopes are

$$\boxed{\underset{k\times 1}{\hat{\beta}} = \begin{bmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (X'X)^{-1}X'y}$$

with **variance-covariance matrix**

$$\boxed{\underset{k\times k}{V(\hat{\beta}|X)} = \begin{bmatrix} \text{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \text{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2(X'X)^{-1}}.$$

Is $\hat{\beta}$ reliable?

Is $\hat{\beta}$ a reliable estimator?

- **Unbiased?**
  $\rightarrow$ Will estimate $\hat{\beta}$ be on average like true β?
  $\rightarrow$ Ideally: $\mathbb{E}(\hat{\beta}) = \beta$
- **Efficient?**
  $\rightarrow$ Will estimate $\hat{\beta}$ be usually close to β?
  $\rightarrow$ Ideally: small var$(\hat{\beta})$

Bias: does $E(\hat{\beta}|X) = \beta$ hold?

Is our set of estimates $\hat{\beta}_1 \ldots \hat{\beta}_k$ unbiased?

$$\begin{aligned}
\hat{\beta} &= (X'X)^{-1}X'y \\
&= (X'X)^{-1}X'(X\beta + u) && \text{given } \boxed{y = X\beta + u} \\
&= \beta + (X'X)^{-1}X'u \\
\mathbb{E}(\hat{\beta}|X) &= \beta + \mathbb{E}[(X'X)^{-1}X'u|X] \\
&= \beta + (X'X)^{-1}X'\mathbb{E}[u|X] \\
&= \beta && \text{given } \boxed{\mathbb{E}[u|X] = 0}
\end{aligned}$$

Ubiasedness may not hold if:

- true data generating process differs from the estimated model $y = X\beta + u$
- $\text{cov}(x_j, u) \neq 0$, for some $j \in \{1, \ldots, k\}$, violating assumption $\mathbb{E}[u|X] = 0$

Bias: does $E(\hat{\beta}|X) = \beta$ hold?

Thus, unbiasedness relies on key assumptions:

1. **Correct specification**. Unobserved reality behaves like $\boxed{y = X\beta + u}$.
2. **Exogenous errors**. The error has zero mean given *any* value of X: $\boxed{\mathbb{E}(u|X) = 0}$. This implies $\text{cov}(x_j, u) = 0$ for all $j = 1 \ldots k$.

When is the variance low?

We obtained the variance-covariance matrix

$$\underset{k\times k}{V(\hat{\beta}|X)} = \begin{bmatrix} \mathrm{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \mathrm{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2(X'X)^{-1}$$

For $k = 2$ (univariate regression), we know that the slope variance is

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2} = \frac{\sigma^2}{n\ \text{var}(x)}$$

which declines with $\sigma^2 \downarrow$ and $n \uparrow$.

Key questions:

- What's the variance of any coefficient $\hat{\beta}_j$ when $k > 2$?
- Does $\mathrm{var}(\hat{\beta}_j|X)$ change as we include/exclude other variables?

Frisch-Waugh-Lovell (FWL) Theorem

Frisch-Waugh-Lovell (FWL) Theorem

The **Frisch-Waugh-Lovell (FWL) Theorem** provides an extremely useful tool to study $\mathbb{E}(\hat{\beta}_j)$ and $\mathrm{var}(\hat{\beta}_j)$ as a function of other included variables.

Key ingredients

- Auxiliary regression of x_j w.r.t. all other regressors $x_{\neq j}$

$$x_j = X_{\neq j}\hat{\gamma} + \underbrace{\tilde{x}_j}_{\text{residuals}}$$

- Residuals $\tilde{x}_j$ of that auxiliary regression
- Fit-goodness R_j^2 of that of that auxiliary regression

Frisch-Waugh-Lovell (FWL) Theorem

Theorem claim 1:

Estimating the OLS coefficient $\hat{\beta}_j$ through the multiple regression

$$y = X\hat{\beta} + \hat{u},$$

or the the OLS coefficient $\tilde{\beta}_j$ through the simple regression

$$\hat{y} = \tilde{\beta}_1 + \tilde{\beta}_j \tilde{x}_j + \tilde{u},$$

(with $\tilde{x}_j =$ residuals from auxiliary regression) yields the same

$$\boxed{\tilde{\beta}_j = \hat{\beta}_j} \text{ and } \boxed{\mathrm{var}(\tilde{\beta}_j) = \mathrm{var}(\hat{\beta}_j)}.$$

Frisch-Waugh-Lovell (FWL) Theorem

Example:

	Dependent variable:		
	income2010 (1)	ineq1950 (2)	income2010 (3)
ineq1950	−16,181.260*** (4,466.854)		
income1950	0.844 (0.552)	−0.0001 (0.00003)	
xtilde			−16,181.260** (6,546.118)
Constant	49,298.500*** (10,741.530)	2.330*** (0.179)	20,239.130*** (1,809.628)
Observations	13	13	13

Note equal estimate $-16, 181.260$. Warning: The reported standard errors in (3) are misleading; in practice real variance of the coefficient in (1) and (3) must obviously coincide.

Frisch-Waugh-Lovell (FWL) Theorem

Theorem claim 2:

The variance of any regressor $j \in \{1, \dots, k\}$ is

$$\boxed{\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2}}$$

where R_j^2 is the R^2 of the auxiliary regression of x_j against all other regressors.

Where does claim 2 come from?

Consider: $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + u$

Auxiliary regression: $\underbrace{x_2}_{\text{Dep. var}} = \delta_1 + \delta_2 x_3 + \underbrace{\tilde{x}_2}_{\text{residual}}$ yields

$$R^2_{\text{aux}} = 1 - \frac{\text{var}(\tilde{x}_2)}{\text{var}(x_2)}, \text{ hence } \text{var}(\tilde{x}_2) = (1 - R^2_{\text{aux}})\,\text{var}(x_2)$$

Following claim 1 of the FWL theorem

$$\hat{\beta}_2 = \tilde{\beta}_2 = \frac{\text{cov}(\tilde{x}_2, y)}{\text{var}(\tilde{x}_2)}$$

Thus:

$$\text{var}(\hat{\beta}_2|X) = \frac{\sigma^2}{n\,\text{var}(\tilde{x})} = \frac{\sigma^2}{n(1 - R^2_{\text{aux}})\,\text{var}(\tilde{x})} \quad \text{<- var}(\text{x}_\text{j})$$

More generally

For x_j with k total regressors:

- Regress x_j on all other x's:
  $$x_j = \delta_0 + \delta_1 x_1 + \cdots + \delta_{j-1} x_{j-1} + \delta_{j+1} x_{j+1} + \cdots + \delta_k x_k + \underbrace{\tilde{x}_j}_{\text{residual}}$$
- Get R_j^2 from this regression
- Residual variance: $\sum \tilde{x}_{ji}^2 = (1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2$

Thus:

$$\boxed{\mathrm{var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2}}$$

Interpretation

Rewriting in compact notation:

$$\mathrm{var}(\hat{\beta}_j) = \frac{\sigma^2}{n} \cdot \frac{1}{1 - R_j^2} \cdot \frac{1}{\mathrm{var}(x_j)}$$

Where:

- $\frac{\sigma^2}{n}$: Basic sampling variance
- $\frac{1}{\mathrm{var}(x_j)}$: More variable $x \Rightarrow$ more precision
- $\frac{1}{1-R_j^2}$: **Variance Inflation Factor (VIF)**

Interpretation: When x_j is highly predictable from other x's ($R_j^2 \to 1$), VIF $\to \infty$ and variance explodes!

Omitted Relevant Variables

Population Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad (E[\varepsilon|x_1, x_2] = 0)$$

Estimated Model:

$$y = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{u}$$

Problem: Now our error term is $u = \beta_2 x_2 + \varepsilon$!!!

$\rightarrow$ Exogeneity assumption cov$(x_1, u) = 0$ invalidated when:

- $\beta_2 \neq 0$ (causal effect exists)
- cov$(x_1, x_2) \neq 0$ (variables correlated)

Omitted variable bias (OVB)

Consider the case $x_2 = \delta x_1 + \nu$.

We estimate

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\text{cov}(x_1, y)}{\text{var}(x_1)} \\
&= \frac{\text{cov}(x_1, \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon)}{\text{var}(x_1)} \\
&= \beta_1 + \frac{\text{cov}(x_1, \beta_2 x_2 + \varepsilon)}{\text{var}(x_1)} \\
&= \beta_1 + \beta_2 \frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)} + \frac{\text{cov}(x_1, \varepsilon)}{\text{var}(x_1)} \\
E[\hat{\beta}_1|X] &= \beta_1 + \beta_2 \delta
\end{aligned}$$

Omitted variable bias (OVB)

Omitted Variable Bias

- True model: $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$
- Estimated model: $y = \hat{\beta}_1 x_1 + \hat{u}$
- Bias: $E[\hat{\beta}_1] = \beta_1 + \beta_2 \delta$
  where δ from $x_2 = \delta x_1 + v$
- Bias occurs when:
  1. $\beta_2 \neq 0$ (omitted variable matters)
  2. $\delta \neq 0$ (x_2 correlates with x_1)

Example:Light Exposure and Myopia Development

Conflicting Studies:

- **UPenn (1999, Nature):** Artificial light exposure ↑ myopia risk in infants
- **Ohio State (2000, Nature):** No significant effect of light exposure

Variable Mapping:

- y: Future myopia development
- X_1: Light exposure + controls
- X_2: Parental myopia (omitted in UPenn study)
- $\beta_2 > 0$: Genetic transmission exists
- $\delta > 0$: Myopic parents tend to leave lights on

Example:Light Exposure and Myopia Development

OV-Bias

$$E[\hat{\beta}_{light}] = \beta_{light} + \underbrace{\delta \cdot \beta_2}_{Bias>0}$$

- $\beta_2 > 0$ (genetic transmission exists)
- $\delta > 0$ (myopic parents use more light)
- $\Rightarrow$ Positive bias in light exposure coefficient

Consequences:

- Overestimated t-statistic: $t_{light} = \frac{\hat{\beta}_{light}}{\sqrt{\widehat{var}}}$
- Underestimated p-value
- Exaggerated significance
- UPenn: Found spurious significance due to omitted variable bias

Parental myopia was the confounding variable explaining both light exposure patterns and child myopia.

Variance Analysis

Variance (by FWL):

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{n} \frac{1}{(1 - R^2_{x_1,x_2})} \frac{1}{\text{var}(x_1)}$$

where:

- $R^2_{x_1,x_2}$ from regressing x_1 on x_2
- σ^2 estimated with $\hat{\sigma}^2 = \frac{SSR}{n-k}$

Key Insight: Omitting x_2:

- Reduces denominator $(1 - R^2) \downarrow$ (lowers variance)
- But introduces bias $\beta_2\delta$

Included Irrelevant Variables

Population Model:

$$y = \beta_1 x_1 + u \quad (E[u|x_1] = 0)$$

Estimated Model:

$$y = \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{u}$$

Now cov$(u, x_1) = 0$ holds!
Is this better? Should we try to include as many variables as possible?

Bias & Variance

Unbiasedness Preservation (since now cov$(u, x) = 0$ holds)

$$E[\hat{\beta}_1] = \beta_1 \quad \text{(when } E[u|x_1, x_2] = 0\text{)}$$

Variance Inflation:

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{n} \frac{1}{(1 - R^2_{x_1,x_2})} \frac{1}{\text{var}(x_1)}$$

- $R^2_{x_1,x_2} > 0$ inflates variance via VIF
- And since σ^2 must be estimated via $\hat{\sigma}^2 = \sum \hat{u}_i^2/(n-k)$: No. of regressors $k \uparrow \rightarrow \hat{\sigma}^2 \uparrow$, especially when n is low.

Redundancy & Variance Inflation

- VIF formula: $VIF = \frac{1}{1-R^2_{x_j|x_{-j}}}$
  where $R^2_{x_j|x_{-j}}$ comes from regressing x_j on other predictors
- Interpretation:
  - $R^2 = 0.75 \Rightarrow VIF = 4$ (4× variance inflation)
  - $R^2 = 0.99 \Rightarrow VIF = 100$ (severe multicollinearity)

Problem: Including irrelevant x_2 when:

- $\beta_2 = 0$ (no causal effect)
- cov$(x_1, x_2) \neq 0$ (variables correlated)

What VIF Measures

Auxiliary regression: Regress x_1 on all other predictors $(x_2, x_3, \dots)$

- R^2_{aux} close to 0: Other predictors don't explain x_1 well
  - VIF ≈ 1 (no inflation)
  - Variance stays small
- R^2_{aux} close to 1: Other predictors explain x_1 very well
  - VIF becomes large
  - Variance gets inflated

PRACTICAL IMPLICATIONS

Bias-Variance Tradeoff

- Omitted variables:
  Bias: $\beta_2\delta$
  Variance: May $\downarrow$ (if x_2 noisy) or $\uparrow$ (if x_2 informative)
- Redundant variables:
  Bias: None
  Variance: $\uparrow$ via VIF
- Practical dilemma:
  - Overspecification $\Rightarrow$ efficiency loss
  - Underspecification $\Rightarrow$ potential bias
  - Solution: Think about really relevant variables. **Avoid redundancy** but **include possible confounders** of the treatment.

Bias-Variance Tradeoff:

	Omitted Relevant	Included Irrelevant
Bias	$\beta_2\delta \neq 0$	0
Variance	Potentially $\downarrow$	$\uparrow$ via VIF
Efficiency	May improve	Always worse

Practical examples

Consider the following data generating process

```
set.seed(123) # optional, for exact reproducibility

# Parameters to play with
n <- 50           # Number of observations
sigma2 <- 1       # Error variance
beta1 <- 2        # True coefficient for x1
beta2 <- -2       # True coefficient for x2 (set to 0 for irrelevant variable)
delta <- 0.9      # Relationship between x1 and x2 (cov(x1,x2)/var(x1))

# Generate data
x1 <- rnorm(n)
x2 <- delta*x1 + rnorm(n, sd = sqrt(1 - delta^2))  # Ensure var(x2)=1
u <- rnorm(n, sd = sqrt(sigma2))
y <- beta1*x1 + beta2*x2 + u
irrelevant <- 1.2*x1 + .1*rnorm(n) # irrelevant but correlated with x1
```

Let the "unknown" parameter of interest be $\boxed{\beta_1 = 2}$.
Notice that the irrelevant variable is correlated with x_1

Practical examples

	Omitted (1)	Correct (2)	Overfit (3)
x1	0.262	2.375***	2.769
	(0.212)	(0.355)	(1.961)
x2		−2.389***	−2.401***
		(0.361)	(0.370)
Irrelevant			−0.323
			(1.579)
Observations	50	50	50
R2	0.031	0.498	0.498
Note:			*p<0.1; **p<0.05; ***p<0.01

Results:

- In (1), omitting x_2 biases $\hat{\beta}_1 \downarrow$ (via $\delta\beta_2$). (2) performs better.
- In (3): VIF$\uparrow \rightarrow$ var$(\hat{\beta}_1) \uparrow \rightarrow \hat{\beta}_1$ not statistically significant.

Practical examples

Now change to

```
beta2 <- -0.02  # True coefficient for x2 (set to 0 for irrelevant variable)
```

where β_2 continues to be relevant bus has a lower impact on the outcome y ($\beta_2 = -0.02$).

Practical examples

	Omitted (1)	Correct (2)	Overfit (3)
x1	2.013***	2.375***	2.769
	(0.154)	(0.355)	(1.961)
x2		−0.409	−0.421
		(0.361)	(0.370)
Irrelevant			−0.323
			(1.579)
Observations	50	50	50
R2	0.780	0.786	0.786
Note:	*p<0.1; **p<0.05; ***p<0.01		

Despite being incorrect, the **model with the omitted variable performs better**!

Why? You should be able to answer ;-)

Diagnosis

The Adjusted R^2

Recall the definition of the R^2

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Because R^2 is a function of $\sum_{i=1}^n \hat{u}_i^2$, we have:

$$\boxed{R^2 \underset{k\to\infty}{\longrightarrow} 1}$$

We cannot use R^2 to compare the goodness of fit of models with different numbers of regressors, due to a systematic bias where models with high k tend to have higher R^2.

The Adjusted R^2

A correction for the bias introduced by k is to normalize the coefficient by the degrees of freedom of $\sum_{i=1}^{n} \hat{u}_i^2$ and $\sum_{i=1}^{n}(y_i - \bar{y})^2$, which are $n-k$ and $n-1$ respectively. The resulting statistic is called the **adjusted coefficient of determination** or corrected R^2:

$$
\begin{aligned}
\bar{R}^2 &= 1 - \frac{\sum_{i=1}^{n} \hat{u}_i^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \frac{(n-1)}{(n-k)} \\
&= 1 - (1 - R^2)\frac{n-1}{n-k}
\end{aligned}
$$

Application to polynomial regression

'Unknown' data generating process:

```
# Define coefficients
b1 <- 2
b2 <- 1.5
b3 <- -0.05
b4 <- 0.0005
b5 <- 0.0002

# Generate random uniform x values (between 10 and 100)
z <- runif(n, 10, 100)

# Data generating process (unknown to the researcher)
y4 <- b1 + b2*z + b3*z^2 + b4*z^3 + b5*z^4 + rnorm(n, 0, 200)
```

Application to polynomial regression

Then 'naively' run polynomial regression up to order 7

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
z1	187.3132***	-258.8695***	92.8726***	-15.9799	-40.9772	49.4034	474.7408
	(10.0725)	(11.7546)	(13.4164)	(38.0644)	(88.0659)	(210.4774)	(483.2892)
z2		4.0506***	-3.2942***	0.3900	1.6062	-4.1634	-37.5053
		(0.1043)	(0.2699)	(1.2397)	(4.0556)	(12.8549)	(36.4448)
z3			0.0444***	-0.0043	-0.0306	0.1449	1.4582
			(0.0016)	(0.0161)	(0.0851)	(0.3808)	(1.3961)
z4				0.0002***	0.0005	-0.0023	-0.0308
				(0.0001)	(0.0008)	(0.0059)	(0.0297)
z5					-0.000001	0.00002	0.0004
					(0.000003)	(0.00004)	(0.0004)
z6						-0.000000	-0.000002
						(0.000000)	(0.000002)
z7							0.0000
							(0.0000)
Constant	-5,965.6510***	3,721.8920***	-758.4058***	243.0720	420.7805	-98.3478	-2,176.1190
	(614.6885)	(292.0271)	(190.7420)	(376.9863)	(679.3492)	(1,291.8580)	(2,487.1100)
Observations	100	100	100	100	100	100	100
R2	0.7792	0.9867	0.9985	0.9986	0.9986	0.9986	0.9986
Adjusted R2	0.7769	0.9864	0.9984	0.9986	0.9986	0.9985	0.9985

Application to polynomial regression

Notice that $\bar{R}^2$ begins to fall after order 4.

Alternative intra-sample indicators that you may learn in Econometrics II:

- AIC (Akaike Information criterion)
- BIC (Bayes Information criterion)
- Hannan-Quinn criterion

Ideally: Out-of-Sample Evaluation

The ideal criterion to judge model quality is to examine its performance "out-of-sample" using data **not used** for estimation.

Procedure:

1. Estimate models with n observations
2. Evaluate performance with n_p prediction data points

Requirement:

1. Large total no. of observations: $N = n + n_p$

Ideally: Out-of-Sample Evaluation

We examine prediction errors $e_i = y_i - \hat{y}_i$. Common evaluation metrics:

1. **Mean Squared Error** (MSE) / RMSE:

$$\textbf{RMSE} = \left(\frac{1}{n_p} \sum^{n_p} (y_i - \hat{y}_i)^2 \right)^{\frac{1}{2}}$$

2. **Mean Absolute Error** (MAE):

$$\textbf{MAE} = \frac{1}{n_p} \sum^{n_p} |y_i - \hat{y}_i|$$

Where n_p = number of predicted observations.

Takeaways

Key takeaways

You should now know:

- What is the omitted variable bias (OVB)
- What the OVB means for the assumption $\mathbb{E}[u_i|X] = 0$
- The impact of the OVB on $\mathbb{E}(\hat{\beta}_j|X)$ and $\text{var}(\hat{\beta}_j|X)$
- The consequences of including irrelevant variables
- The nature of the bias-efficiency tradeoff
- The FWL theorem
- Diagnosis through intra-sample (Adjusted-R^2) or out-of-sample (e.g. MSEP) indicators.